Omiprep Data Preparation Summary Report

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1 Project Information

This `omiprep` report summarizes the data preparation steps for:

- **Project:** fom1

1.1 Overview

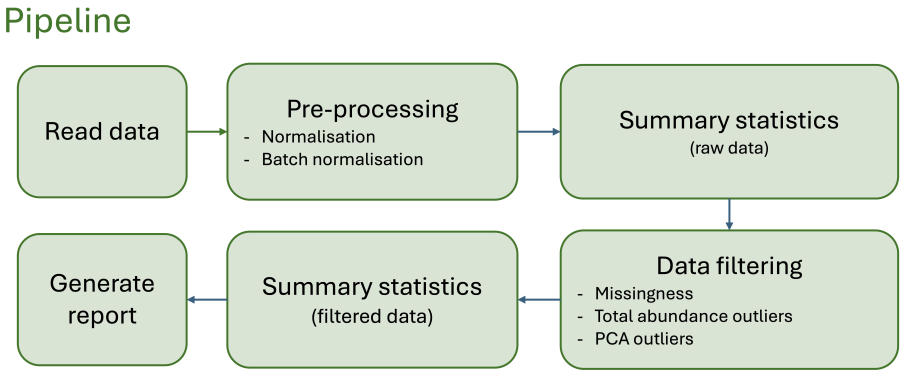
The `omiprep` R package performs three key operations:

1. **Assessment & Summary Statistics**: Provides an initial assessment of raw 'omics data.
2. **Data Filtering**: Applies filtering techniques to clean the dataset.
3. **Post-Filtering Assessment**: Evaluates the filtered dataset, particularly in relation to batch variables when available.

This report contains descriptive information on both raw and filtered 'omics data for **fom1**.

Please raise any issues on the GitHub issues page.

1.2 Data preparation workflow:



2 Summary of raw data

2.1 Sample size of fom1 data set:

Table 1: Sample Size Summary

Dataset	Samples	Features
INPUT	4525	249
QC INCLUDED	4512	249

3 Missingness

Missingness is evaluated across samples and features using the original (input) data set.

3.1 Visual structure of missingness in the raw data



3.2 Summary of sample and feature missingness

The extent of missing data in both samples and features of the dataset was evaluated - for each sample, the proportion of features with missing values, and for each feature, the proportion of samples with missing values. This can identify whether missingness is more strongly driven by poorly measured features or by problematic samples.

The distribution of missingness is summarized in:

1. histograms
2. tabulated using percentiles, and
3. estimates of sample and feature samples sizes under various levels of missingness filtering thresholds

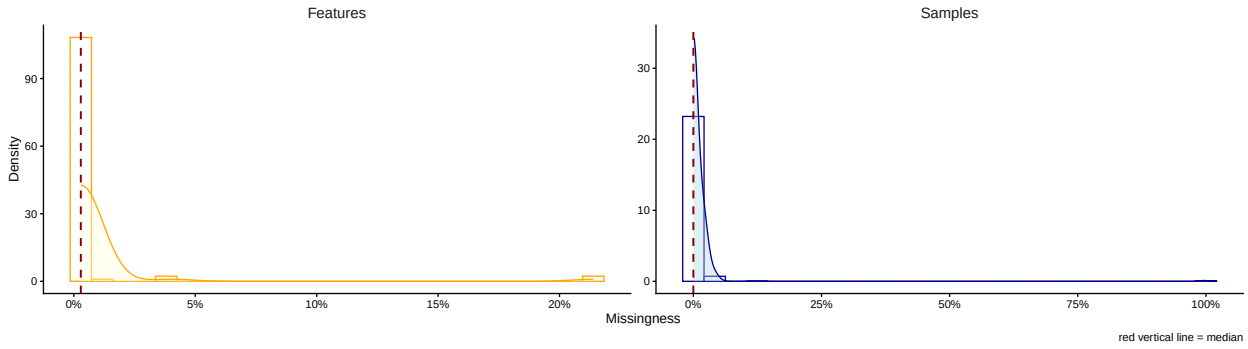


Table 2: Sample and feature missingness percentiles.

	Percentile	Features	Samples
0%	0.00	0.003	0
25%	0.25	0.003	0
50%	0.50	0.003	0
75%	0.75	0.003	0
100%	1.00	0.214	1

Table 3: Estimates of sample and feature sample sizes at different missingness thresholds.

Missingness.threshold	Number.of.Samples	Number.of.Features
5%	4507	244
10%	4509	244
15%	4512	244
20%	4512	244
25%	4512	249
30%	4512	249
35%	4512	249
40%	4512	249
45%	4512	249
50%	4512	249
55%	4512	249
60%	4512	249
65%	4512	249
70%	4512	249
75%	4512	249
80%	4512	249
85%	4512	249
90%	4512	249
95%	4512	249
100%	4512	249

4 Data Filtering

4.1 Exclusion summary

Table 4: Sample and Feature Exclusions

Reason	Count
Features	
- user excluded	0
- extreme feature missingness	0
- user defined feature missingness	0
- user defined feature skewness	0
Samples	
- user excluded	0
- extreme sample missingness	13
- user defined sample missingness	0
- user defined sample totalpeakarea	0
- user defined sample pca outlier	0

Note:
9 primary data filtering exclusion checks were evaluated during data preparation. User-defined thresholds are indicated with an asterisk.

[1] Features manually excluded.
[2] Features with missingness >=80%.
[3] Features exclusions based on the user defined threshold of *>=50%.
[4] Skewness-based feature exclusions were disabled.
[5] Samples manually excluded.
[6] Samples with missingness >=80%.
[7] Samples exclusions based on user defined threshold of *>=20%.
[8] Samples with a total-sum-abundance that is *>=unknown SD from the mean.
[9] Samples that are *>=5 SD from the mean on principal components 1-to-2.

4.2 'Omics feature reduction and principal components

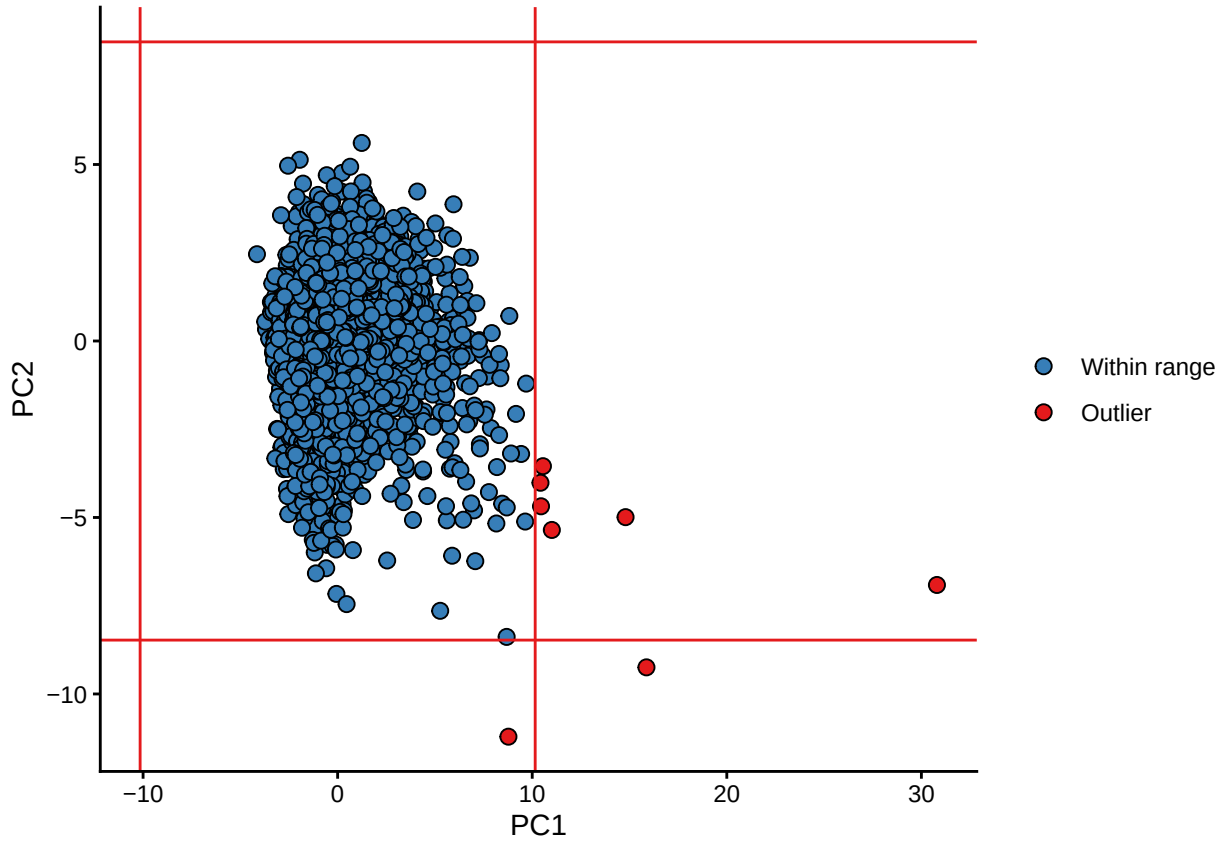
A data reduction was carried out to identify a list of representative features for generating a sample principal component analysis. This step reduces the level of inter-correlation in the data to ensure that the principal components are not driven by groups of correlated features.

The data reduction table presents the number of features at each phase of the data reduction (Spearman's correlation distance tree cutting) analysis.

Table 5: Feature summary

Data reduction	Count
Total feature count	249
Features included in data reduction	244
Number of feature clusters	21
Number of representative features	21

The following plot respresents principal components 1 and 2 using 244 representative features The red vertical and horizontal lines indicate the standard deviation cutoffs for identifying individual outliers. Outliers are those >=5 SD from the mean of PCs 1-2.



5 Summary of filtered data

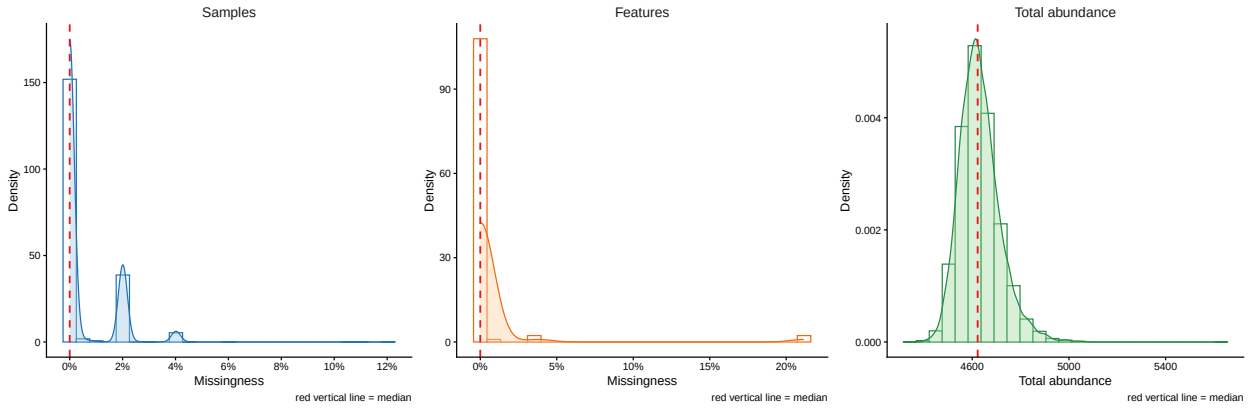
5.1 Sample size (N)

- The number of samples in data = 4512
- The number of features in data = 249

5.2 Relative to the raw data

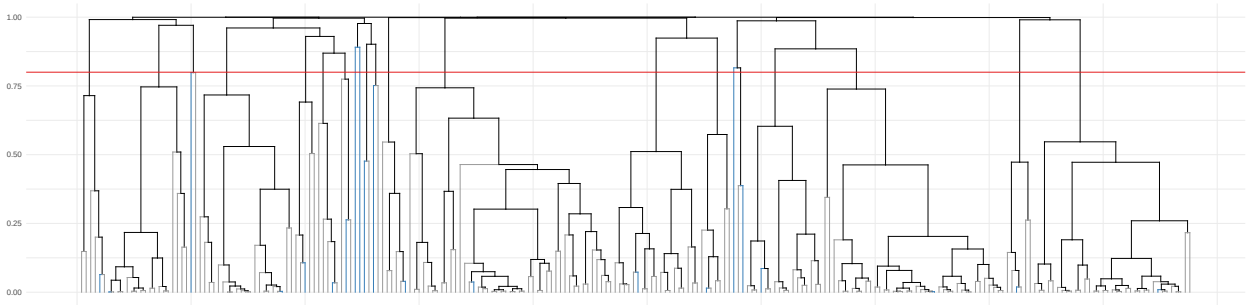
- 13 samples were filtered out, given the user's criteria.
- 0 features were filtered out, given the user's criteria.
- Please review details above and your log file for the number of features and samples excluded and why.

5.2.1 Distributions for sample and feature missingness



5.2.2 Clustering dendrogram of representative features

Spearman's correlation distance clustering dendrogram highlighting the representative features in blue, the clustering tree cut height is denoted by the horizontal line.



5.2.3 Summary of the QC (filtered) 'omics data

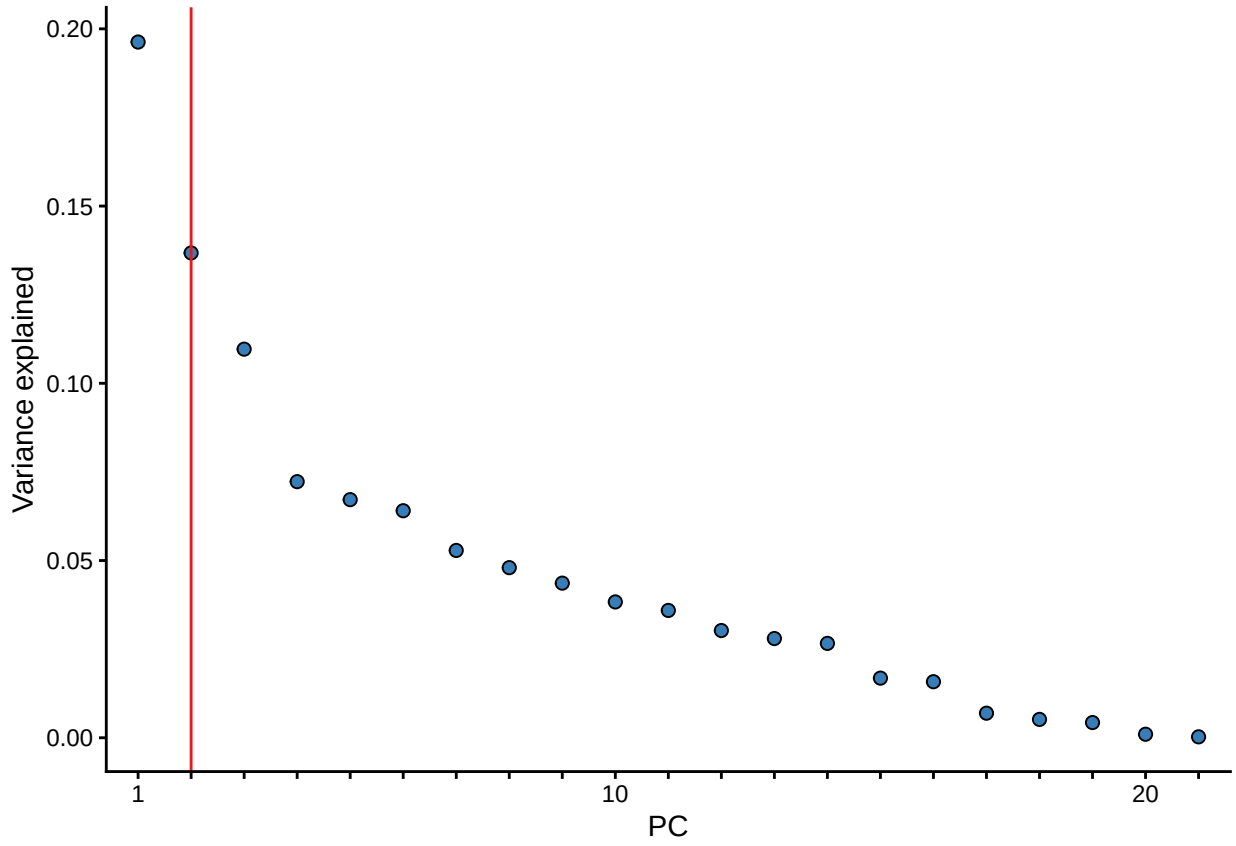
The data reduction table presents the number of features at each phase of the data reduction (Spearman's correlation distance tree cutting) analysis.

Table 6: QC Feature Summary

	Count
Total feature count	249
Features included in data reduction	244
Number of feature clusters	21
Number of representative features	21

5.2.4 Scree plot

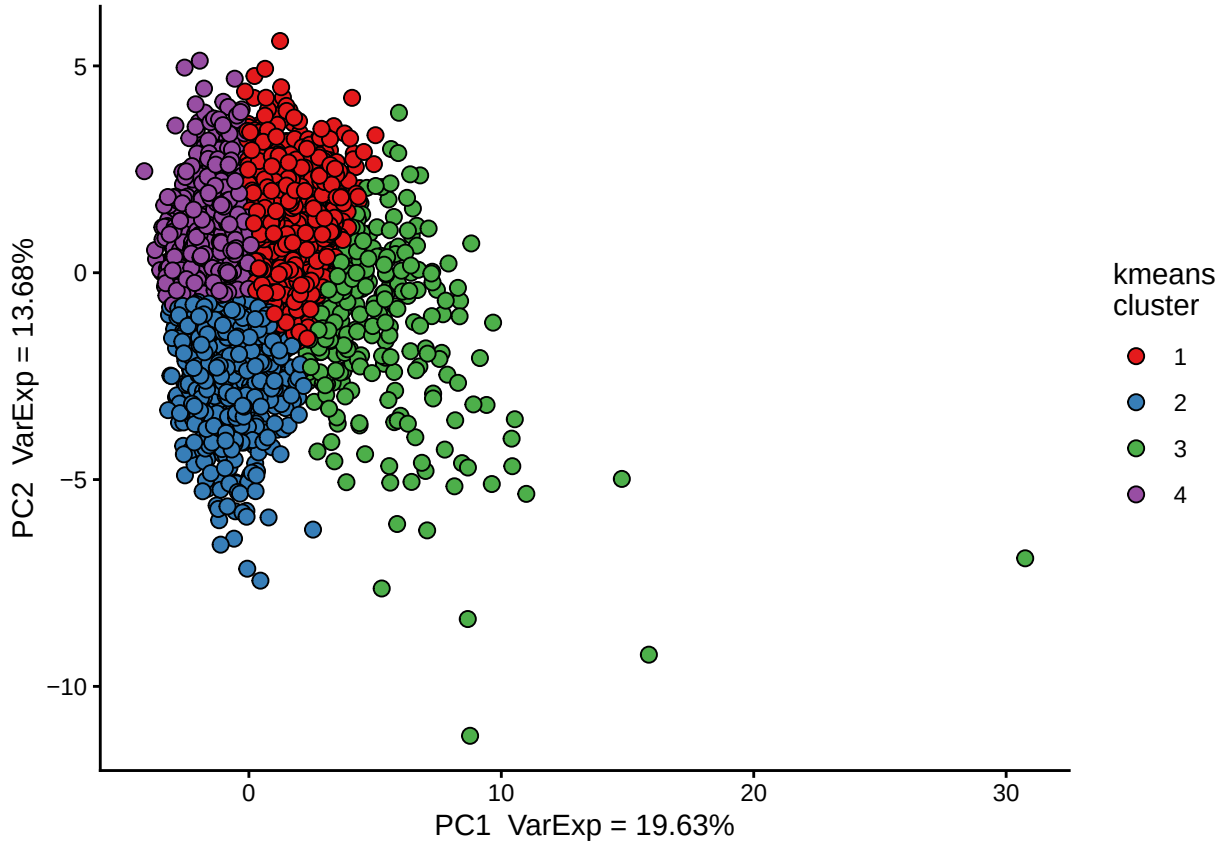
Scree plot of the variance explained by each PC (limited to 100 for plotting) and a plot of principal component 1 and 2, as derived from the representative features The Scree plot also identifies the number of PCs estimated to be informative (vertical lines) by the Cattel's Scree Test acceleration factor (red, n = 2).



5.2.5 PC plot

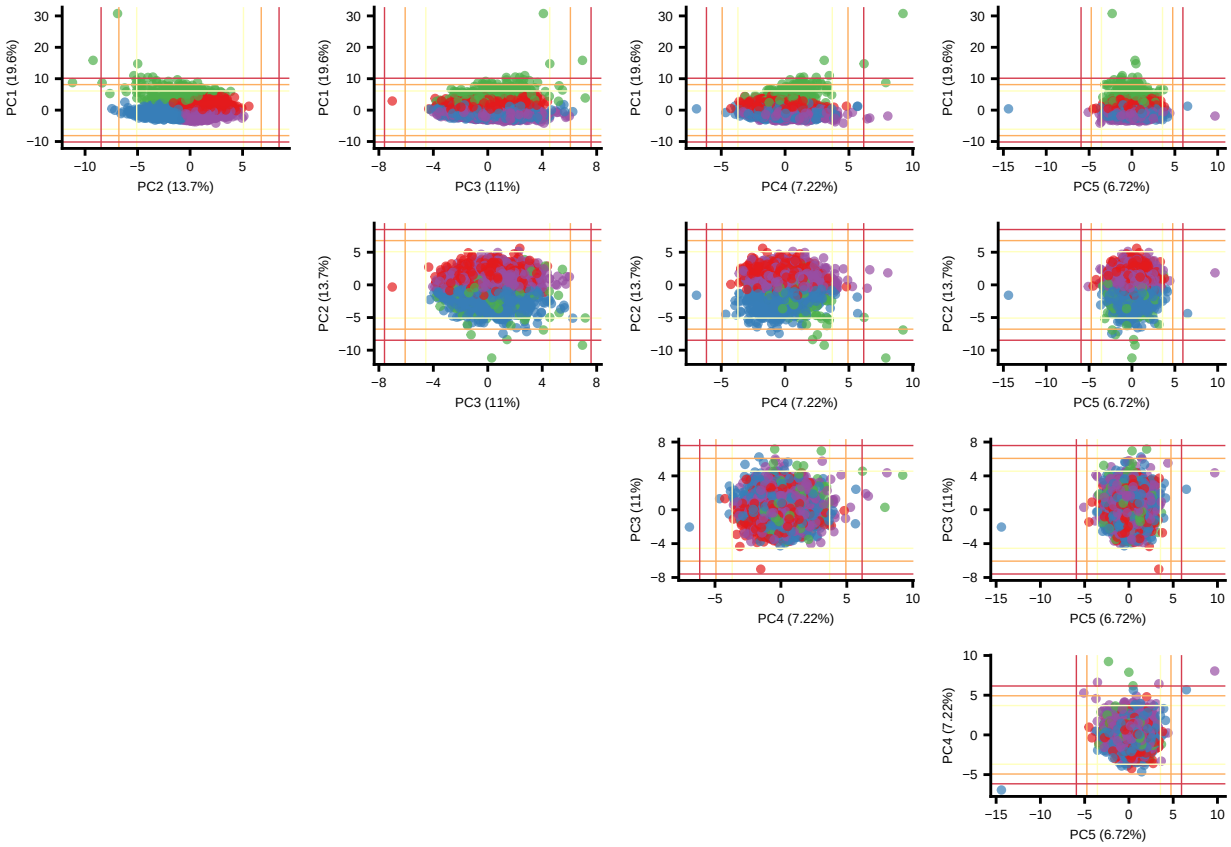
Individuals in the PC plot were clustered into 4 kmeans (k) clusters, using data from PC1 and PC2. The kmeans clustering and color coding is strictly there to help provide some visualization of the major axes of variation in the sample population(s).

The plot presents principal components 1 & 2 using 21 representative features.



5.3 Structure among samples

A matrix (pairs) plot of the top five principal components including demarcations of the 3rd (yellow), 4th (orange), and 5th (red) standard deviations from the mean. Samples are color coded as in the summary PC plot above using a kmeans analysis of PC1 and PC2 with a k (number of clusters) set at 4. The choice of k = 4 was not robustly chosen it was a choice of simplicity to help aid visualize variation and sample mobility across the PCs.



5.4 Feature Distributions

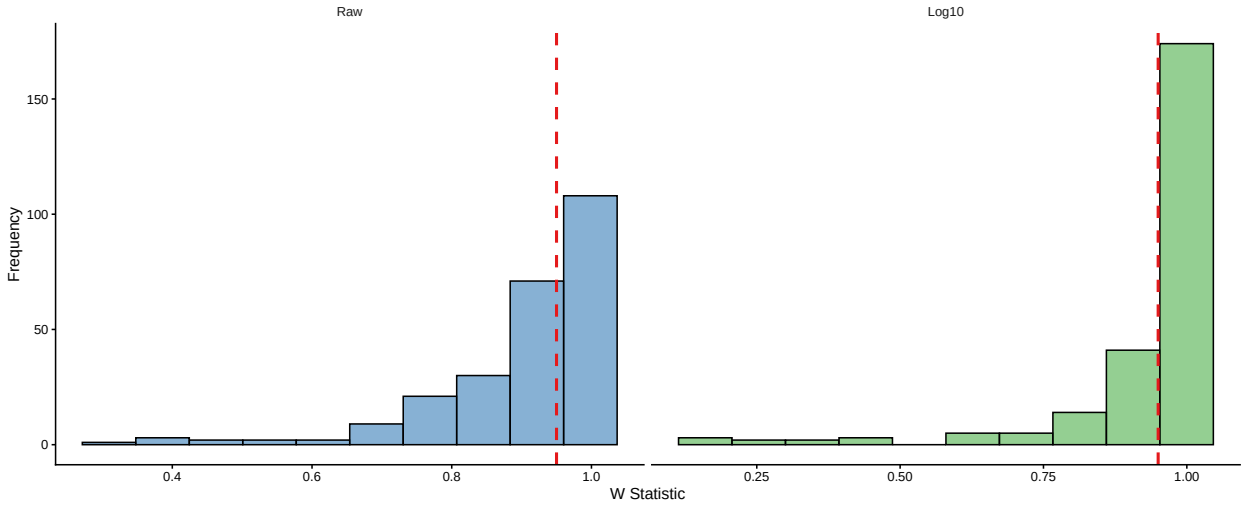
5.4.1 Estimates of normality: W-statistics for raw and log transformed data

Of the 249 features in the data 0 features were excluded from this analysis because of no variation or too few observations (n < 40). Of the remaining 249 features, a total of 130 may be considered normally distributed given a Shapiro W-statistic >= 0.95.

5.4.2 Distribution of W Statistics on Raw and Log10 'Omic Feature Abundances

Histogram plots of Shapiro W-statistics for raw and log transformed data distributions. A W-statistic value of 1 indicates the sample distribution is perfectly normal and value of 0 indicates it is perfectly uniform. Please note that log transformation of the data *may not* improve the normality of your data.

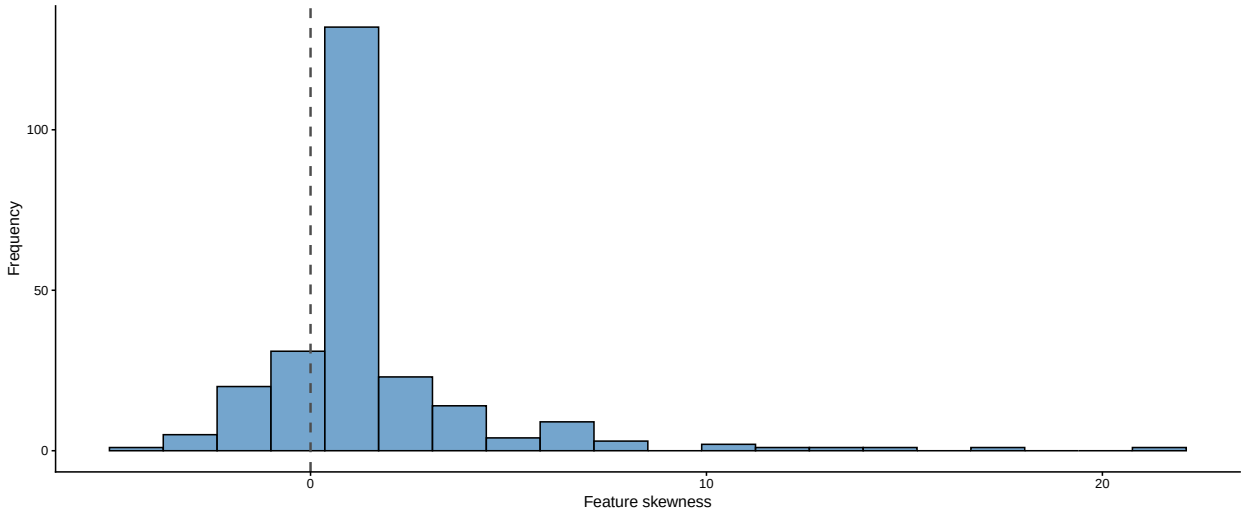
130 of the features exhibit distributions that may declared normal, given a W-stat >= 0.95. In 69 instances (27.7%) of the tested features the log10 data W-stat is < raw data W-stat.



5.4.3 Estimates of skewness on raw abundances

Across the 249 included features, 47 have negative skewness and 202 have positive skewness (mean skewness = 1.44). Strong negative skewness may indicate upper-end assay censoring (an upper detection bound).

No skewness-based feature exclusion threshold was applied during QC.


Frequency
100
50
0
0
10
20
Feature skewness

5.5 Outliers

Evaluation of the number of samples and features that are outliers across the QC data. The below table presents the average number of outlier values for samples and features in the QC data set.

Table 7: Outlier Summary

	Min.	25th	Median	Mean	75th	Max.
Features	0	0	1	8.8110	8	221
Samples	0	0	0	0.4863	0	102

5.5.1 Notes on outlying samples at each feature

There may be extreme outlying observations at individual features that have not been accounted for. You may want to:

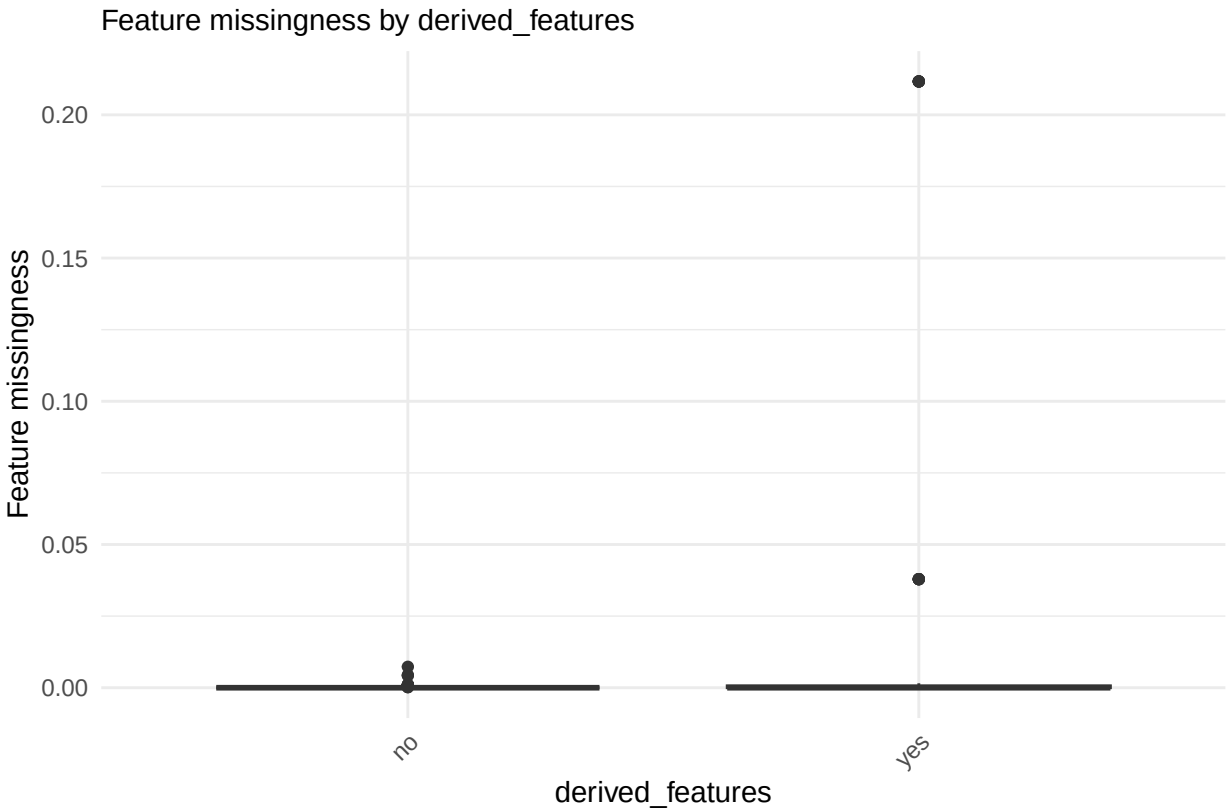
1. Turn these observations into NAs.
2. Winsorize the data to some maximum value.
3. Rank normalize the data which will place those outliers into the top of the ranked standard normal distribution.
4. Turn these observations into NAs and then impute them along with other missing data in your data set.

6 Variation in filtered data by available variables

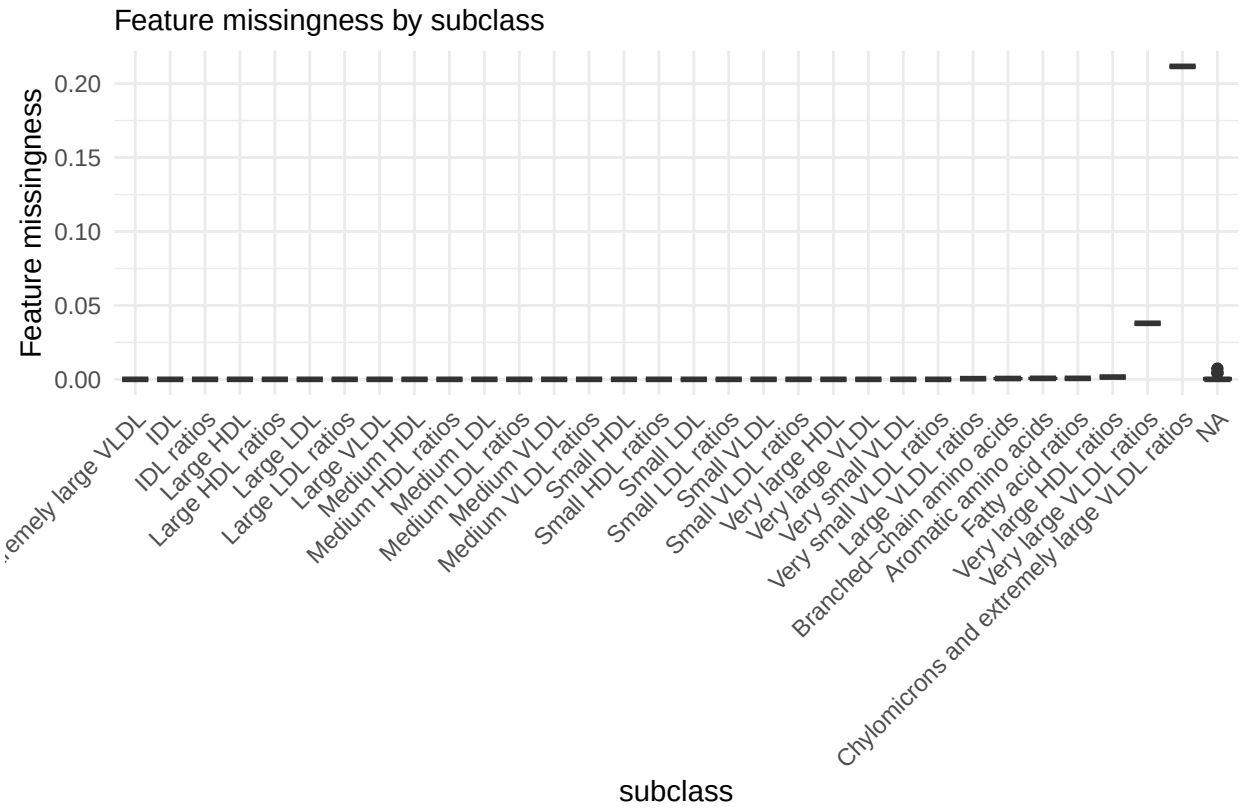
6.1 Feature missingness

Feature missingness may be influenced by the features' biology or pathway classification, or the measurement methodology. The figure(s) below provides an illustrative evaluation of the proportion of *feature missingness* as a product of the variable(s) available in the raw data files.

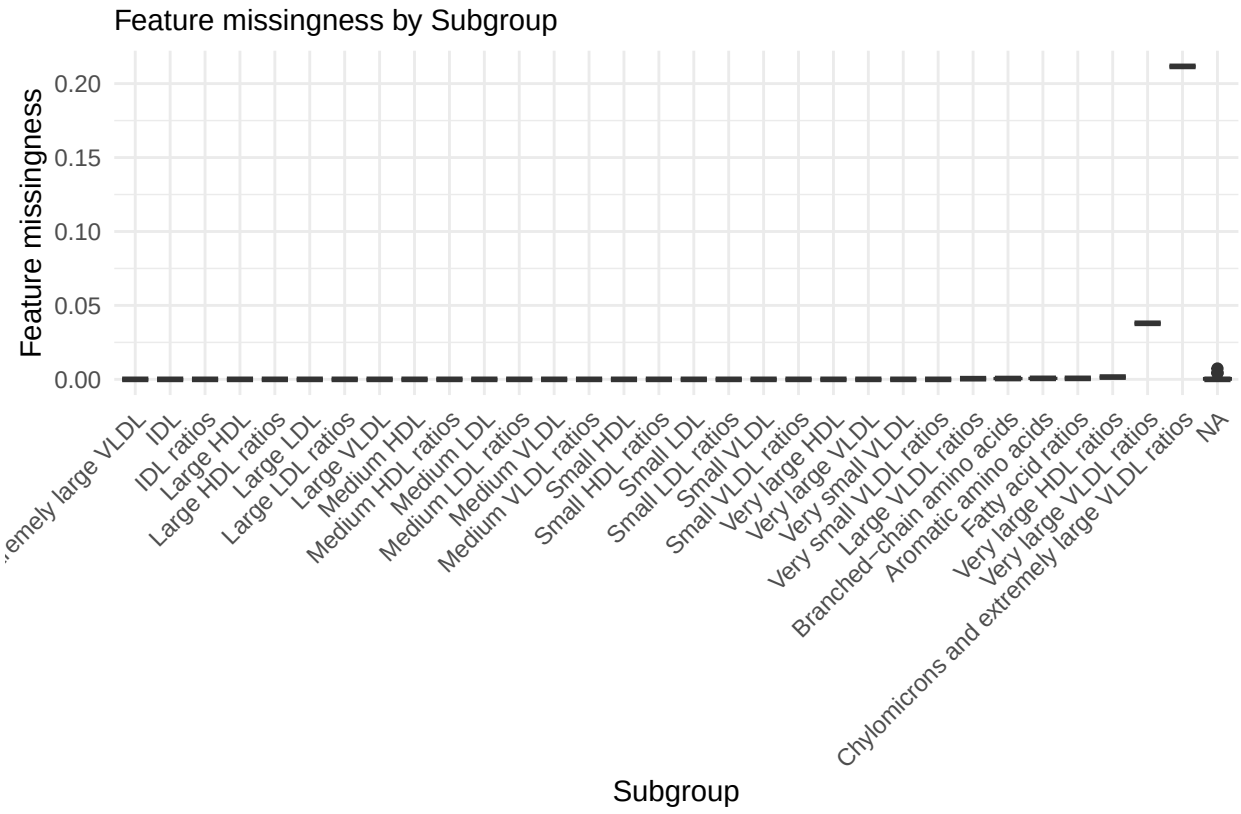
```
-- After filtering a total of 3 feature level batch variables were identified. --
-- They are:
   derived_features
   subclass
   Subgroup
```



5.8% of the variation in Feature missingness can be explained by derived_features (pval = 0.000128)



100% of the variation in Feature missingness can be explained by subclass (pval = 0)



100% of the variation in Feature missingness can be explained by Subgroup (pval = 0)

6.2 Sample missingness

The figure provides an illustrative evaluation of the proportion of *sample missigness* as a product of sample batch variables provided by your supplier. This is the univariate influence of batch effects on *sample missingness*. Box plot illustration(s) of the relationship that available batch variables have with sample missingness.

```
-- No feature level batch variables identified or all were invariable --
```

6.3 Multivariate evaluation: batch variables

TypeII ANOVA: the eta-squared (eta-sq) estimates are an estimation of the percent of variation explained by each independent variable, after accounting for all other variables, as derived from the sum of squares. This is a multivariate evaluation of batch variables on *sample missingness*. Presence of NA's would indicate that the model is inappropriate.

```
[1] " -- No sample level batch variables were provided or all were invariable -- "
```

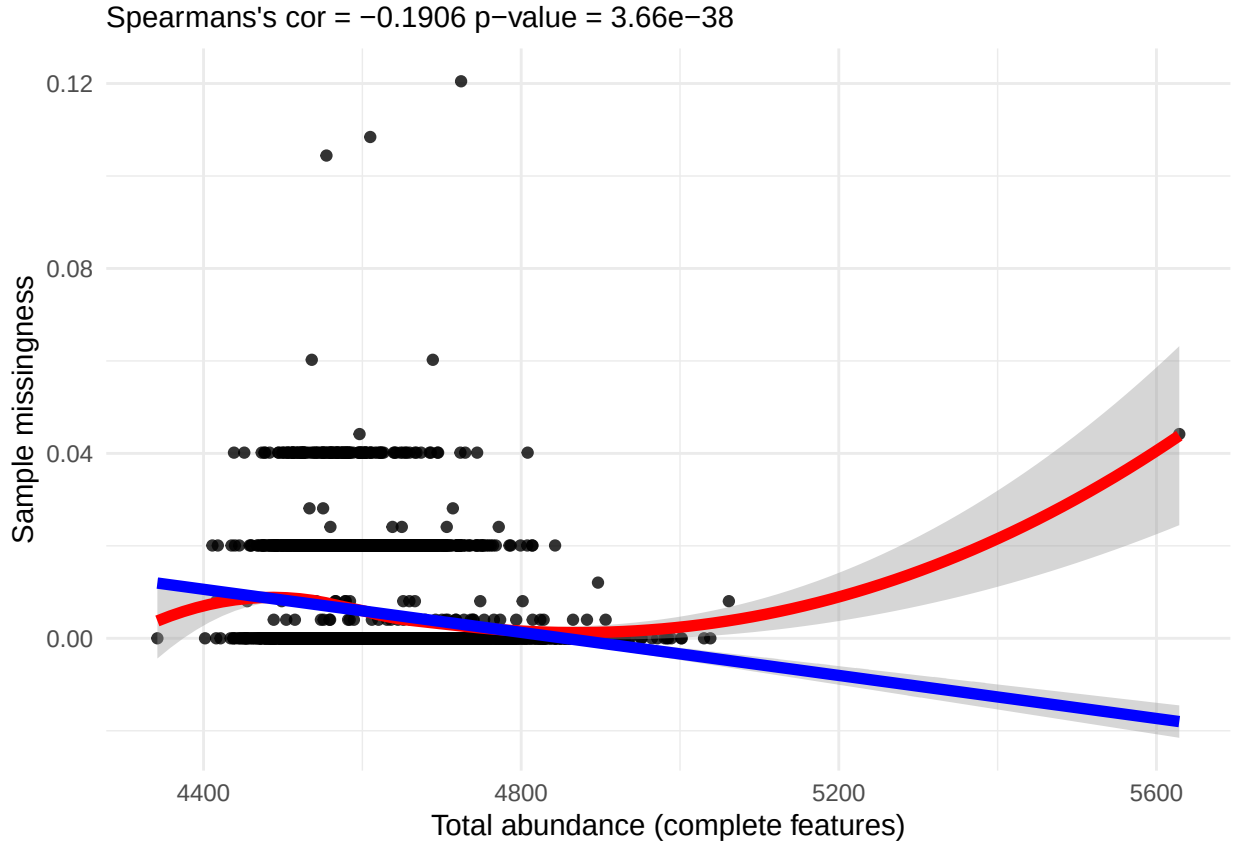
7 Total peak or abundance area (TA) of samples:

The total peak or abundance area (TA) is simply the sum of the abundances measured across all features. TA is one measure that can be used to identify unusual samples given their entire profile. However, the level of missingness in a sample may influence TA. To account for this we:

1. Evaluate the correlation between TA estimates across all features with PA measured using only those features with complete data (no missingness).
2. Determine if the batch effects have a measurable impact on TA.

7.1 Relationship with missingness

Correlation between total abundance (TA; at complete features) and missingness. Relationship between total sum abundance at complete features (x-axis) and sample missingness (y-axis).



7.2 Univariate evaluation: batch effects

The figure below provides an illustrative evaluation of the *total abundance* (at complete features) as a product of sample batch variables provided by your supplier. Violin plot illustration(s) of the relationship between total abundance (TA; at complete features) and sample batch variables that are available in your data.

```
[1] " -- No sample level batch variables were provided or all were invariable -- "
```

7.3 Multivariate evaluation: batch variables

TypeII ANOVA: the eta-squared (eta-sq) estimates are an estimation on the percent of variation explained by each independent variable, after accounting for all other variables, as derived from the sum of squares. This is a multivariate evaluation of batch variables on *total peak|abundance area* at complete features.

```
[1] " -- No sample level batch variables were provided or all were invariable -- "
```

8 Power analysis

Exploration for case/control and continuous outcome data using the filtered data set

Analytical power analysis for both continuous and imbalanced presence/absence correlation analysis.

Simulated effect sizes (standardized by trait SD) are illustrated by their color in each figure. Figure (A) provides estimates of power for continuous traits with the total sample size on the x-axis and the estimated power on the y-axis. Figure (B) provides estimates of power for presence/absence (or binary) traits in an imbalanced design. The estimated power is on the y-axis. The total sample size is set to 99 and the x-axis depicts the number of individuals present (or absent) for the trait. The effects sizes illustrated here were chosen by running an initial set of simulations which identified effects sizes that would span a broad range of power estimates given the sample population's sample size.

